COMPUTER SYSTEMS AND PLATFORM TECHNOLOGIES

IT0047

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Technology Driven by Innovation

MODULE 1

Data Manipulation

MODULE 1.1

Number System - Conversion

At the end of the lesson, the student should be able to :

- Identify the different number system as they relate to computers
- Identify the unit, number, base/radix, positional notation, least significant digit and most significant digit
- Convert values from any number system to each other and back to other systems

Number Base

- What is a **number base**?

  A **number base** is a specific collection of symbols on which a number system can be built.

- The number base familiar to us is *base 10*, upon which the *decimal number system* is built. There are ten symbols - 0 to 9 - used in the decimal system.

Place Value

- What is the concept of **place value**?

  Place value means that the value of a digit in a number depends not only on its own natural value but also on its location in the number.

  It is used interchangeably with the term **positional notation**.

- Place value tells us that the two 4s in the number **3474** have different values, that is, **400**

  and **4**, respectively.

A Review of the Decimal Number System

- The word “decimal” comes from the Latin word *decem*, meaning *ten*.
- Thus, the number base of the decimal number system is *base 10*.
- Since it is in base 10, *ten symbols* are used in the decimal number system.

  {0,1,2,3,4,5,6,7,8,9}
- This means that **only** the digits in the above set can be used for each position in every place

  value in a given decimal number.

A Review of the Decimal Number System

2	7	**0**

- Note that the highlighted place value can be filled by the digits in the set {0,1,2,3,4,5,6,7,8,9}.
- Thus, it can be increased by 1 until it reaches -

2	7	**9**

- At this point, the symbols that can be used to fill the highlighted position has been exhausted. Increasing it further causes a shift in place value and resets the initial place value to zero. Thus -

2	**8**	0

A Review of the Decimal Number System

Case Study: **3474**

Using place values, the number 3474 is understood to mean,

$$3000 + 400 + 70 + 4 = 3474$$

This can also be expressed as –

$$(3x1000) + (4x100) + (7x10) + 4 = 3474$$

Note that each digit is multiplied by **powers of 10**, so that the above is equal to –

$$\mathbf{(3x10^3) + (4x10^2) + (7x10^1) + (4x10^0) = 3474}$$

Note that the rightmost exponent starts from zero and increases by 1 as the place value increases.

Hence, the decimal number system is said to be in ***base 10****.*

Base 2: The Binary Number System

- The word “binary” comes from the Latin word *bis*, meaning *double*.
- Thus, the number base of the binary number system is *base 2*.
- Since it is in base 2, *two symbols* are used in the binary number system.

  {0,1}
- This means that **only** the digits in the above set can be used for each position in every place value in a given binary number.

Base 2: The Binary Number System

1	1	0	**0**

- Note that the highlighted place value can be filled by the digits in the set {0,1}.
- Thus, it can be increased by 1 until it reaches -

1	1	0	**1**

- At this point, the symbols that can be used to fill the highlighted position has been exhausted. Increasing it further causes a shift in place value, and resets the initial place value to zero. Thus -

1	1	**1**	0

Base 2: The Binary Number System

- To avoid confusion, one should write a binary number with *base 2* as its subscript whenever necessary.
- Thus, the binary number **10110** should be written as -

$$\mathbf{10110_2}$$

- It should be read as *"one-zero-one-one-zero base two"* and **NOT** *"ten-thousand one-hundred ten"* since each phrase denotes an entirely different number.

Base 2: The Binary Number System

Case Study: $\mathbf{10110_2}$

- We know that the decimal number 3474 can be expressed as powers of 10 –

$$(3x10^3) + (4x10^2) + (7x10^1) + (4x10^0) = 3474_{10}$$

- In the same manner, the binary number $\mathbf{10110_2}$ can be expressed as **powers of 2** –

$$\mathbf{(1x2^4) + (0x2^3) + (1x2^2) + (1x2^1) + (0x2^0) = 22_{10}}$$

Note that the rightmost exponent starts from zero and increases by 1 as the place value increases.

Hence, the binary number system is said to be in ***base 2****.*

Base 8: The Octal Number System

- The word “octal” comes from the Greek word *oktõ*, meaning *eight*.
- Thus, the number base of the octal number system is *base 8*.
- Since it is in base 8, *eight symbols* are used in the octal number system.

  {0,1,2,3,4,5,6,7}
- This means that **only** the digits in the above set can be used for each position in every place

  value in a given octal number.

Base 8: The Octal Number System

3	6	**0**

- Note that the highlighted place value can be filled by the digits in the set {0,1,2,3,4,5,6,7}.
- Thus, it can be increased by 1 until it reaches -

3	6	**7**

- At this point, the symbols that can be used to fill the highlighted position has been exhausted. Increasing it further causes a shift in place value, and resets the initial place value to zero. Thus -

3	**7**	0

Base 8: The Octal Number System

- To avoid confusion, one should write an octal number with *base 8* as its subscript whenever necessary.
- Thus, the octal number **72143** should be written as -

$$\mathbf{72143_8}$$

- It should be read as *"seven-two-one-four-three base eight"* and **NOT** *"seventy two-thousand one-hundred forty three"* since each phrase denotes an entirely different number.

Base 8: The Octal Number System

Case Study: $\mathbf{72143_8}$

- We know that the decimal number 3474 can be expressed as powers of 10 –

$$(3x10^3) + (4x10^2) + (7x10^1) + (4x10^0) = 3474_{10}$$

- In the same manner, the octal number $\mathbf{72143_8}$ can be expressed as **powers of 8** –

$$\mathbf{(7x8^4) + (2x8^3) + (1x8^2) + (4x8^1) + (3x8^0) = 29795_{10}}$$

Note that the rightmost exponent starts from zero and increases by 1 as the place value increases.

Hence, the octal number system is said to be in ***base 8****.*

Base 16: The Hexadecimal Number System

- The word “hexadecimal” is a combination of the Greek word *hex*, meaning *six* and the Latin word *decem*, meaning *ten*.
- Thus, the number base of the hexadecimal number system is *base 16*.
- Since it is in base 16, *sixteen symbols* are used in the hexadecimal number system.

  {0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}
- This means that **only** the digits in the above set can be used for each position in every place value in a given hexadecimal number.

Base 16: The Hexadecimal Number System

A 3 B **0**

- Note that the highlighted place value can be filled by the digits in the set {0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F}.
- Thus, it can be increased by 1 until it reaches -

A 3 B **F**

- At this point, the symbols that can be used to fill the highlighted position has been exhausted. Increasing it further causes a shift in place value, and resets the initial place value to zero. Thus -

A 3 **C** 0

Base 16: The Hexadecimal Number System

- To avoid confusion, one should write a hexadecimal number with *base 16* as its subscript whenever necessary.
- Thus, the hexadecimal number **B23C** should be written as -

$$\mathbf{B23C_{16}}$$

- It should be read as *"b-two-three-c base sixteen".*

Base 16: The Hexadecimal Number System

Case Study: **$B23C_{16}$**

- We know that the decimal number 3474 can be expressed as powers of 10 –

$$(3x10^3) + (4x10^2) + (7x10^1) + (4x10^0) = 3474_{10}$$

- In the same manner, the hexadecimal number **$B23C_{16}$** can be expressed as **powers of 16** –

$$\mathbf{(11x16^3) + (2x16^2) + (3x16^1) + (12x16^0) = 45628_{10}}$$

Note that the rightmost exponent starts from zero and increases by 1 as the place value increases.

*Hence, the hexadecimal number system is said to be in **base 16**.*

Comparative Values: Bases 10, 2, 8, 16

Base 10	Base 2	Base 8	Base 16
Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	10	8

Comparative Values: Bases 10, 2, 8, 16

Base 10	Base 2	Base 8	Base 16
Decimal	Binary	Octal	Hexadecimal
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F
16	10000	20	10
17	10001	21	11

Conversion from Bases 2, 8, 16 to Base 10

- Technique: *Expand the number using the appropriate base.*
- Convert $\mathbf{11010_2}$ to base 10.

$$(1x2^4) + (1x2^3) + (0x2^2) + (1x2^1) + (0x2^0) = ?$$

$$(1x16) + (1x8) + (0x4) + (1x2) + (0x1) = ?$$

$$16 + 8 + 0 + 2 + 0 = \mathbf{26_{10}}$$

- Convert $\mathbf{1537_8}$ to base 10.

$$(1x8^3) + (5x8^2) + (3x8^1) + (7x8^0) = ?$$

$$(1x512) + (5x64) + (3x8) + (7x1) = ?$$

$$(512) + (320) + (24) + (7) = \mathbf{863_{10}}$$

Conversion from Bases 2, 8, 16 to Base 10

- Technique: *Expand the number using the appropriate base.*

- Convert $\mathbf{AF4_{16}}$ to base 10.

$$(A \times 16^2) + (F \times 16^1) + (4 \times 16^0) = ?$$

$$(10 \times 256) + (15 \times 16) + (4 \times 1) = ?$$

$$2560 + 240 + 4 = \mathbf{2804_{10}}$$

Conversion from Base 10 to Bases 2, 8, 16

- Technique: *Divide the number using the appropriate base, then repeatedly divide the resulting quotients by the same base until it becomes zero. The answer is the combined remainders in reverse order accumulated from the divisions. This technique is called the* ***remainder method****.*
- Convert $\mathbf{133_{10}}$ to base 2.

133 ÷ 2 = 66	r. 1
66 ÷ 2 = 33	r. 0
33 ÷ 2 = 16	r. 1
16 ÷ 2 = 8	r. 0
8 ÷ 2 = 4	r. 0
4 ÷ 2 = 2	r. 0
2 ÷ 2 = 1	r. 0
1 ÷ 2 = **0**	r. 1

Answer: $\mathbf{10000101_2}$

Conversion from Base 10 to Bases 2, 8, 16

- Technique: *Divide the number using the appropriate base, then repeatedly divide the resulting quotients by the same base until it becomes zero. The answer is the combined remainders in reverse order accumulated from the divisions. This technique is called the* ***remainder method****.*
- Convert $\mathbf{7863_{10}}$ to base 8.

7863 ÷ 8 = 982	r. 7
982 ÷ 8 = 122	r. 6
122 ÷ 8 = 15	r. 2
15 ÷ 8 = 1	r. 7
1 ÷ 8 = **0**	r. 1

Answer: $\mathbf{17267_8}$

Conversion from Base 10 to Bases 2, 8, 16

- Technique: *Divide the number using the appropriate base, then repeatedly divide the resulting quotients by the same base until it becomes zero. The answer is the combined remainders in reverse order accumulated from the divisions. This technique is called the* ***remainder method****.*

- Convert $\mathbf{2926_{10}}$ to base 16.

  2926 ÷ 16 = 182 r. 14(E)

  182 ÷ 16 = 11 r. 6

  11 ÷ 16 = **0** r. 11(B)

Answer: $\mathbf{B6E_{16}}$

Conversion from Base 8 to Base 2

- Technique: *Convert each octal digit to a set of three binary digits.*

- Convert **5641_8** to base 2.

5	**6**	**4**	**1**
101	110	100	001

Answer: **101110100001_2**

Conversion from Base 2 to Base 8

- <u>Technique</u>: *Convert each set of three binary digits to an octal digit starting from the right going to the left, adding leading zeroes if necessary.*
- Convert $\mathbf{1111010_2}$ to base 8.

001	**111**	**010**
1	7	2

Answer: $\mathbf{172_8}$

Conversion from Base 16 to Base 2

- Technique: *Convert each hexadecimal digit to a set of four binary digits.*
- Convert **$A3CF_{16}$** to base 2.

A	**3**	**C**	**F**
1010	0011	1100	1111

Answer: **1010001111001111_2**

Conversion from Base 2 to Base 16

- Technique: *Convert each set of four binary digits to a hexadecimal digit starting from the right going to the left, adding leading zeroes if necessary.*

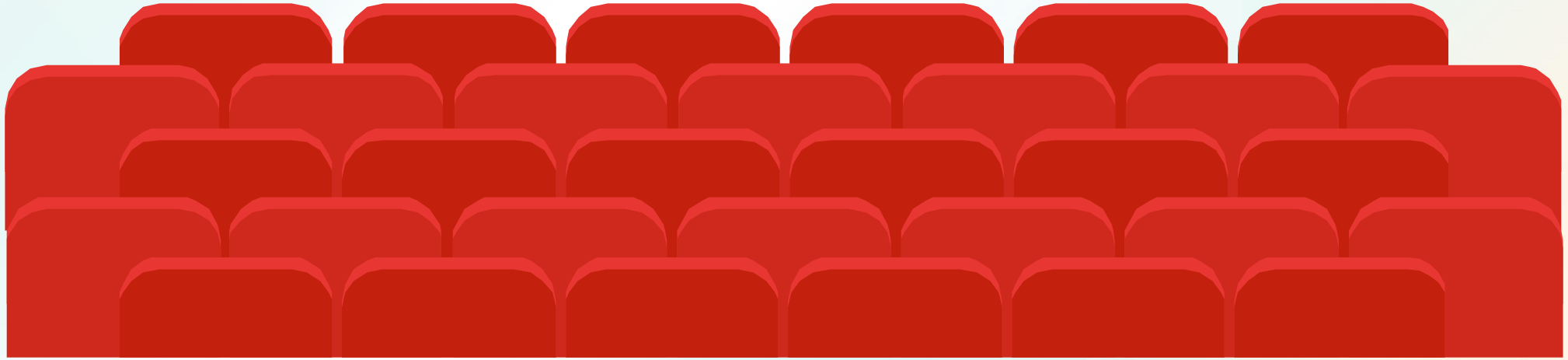
- Convert **101011010_2** to base 16.

0001 0101 1010

1 5 A

Answer: **$15A_{16}$**

Q&A SESSION

ASK ANY QUESTION RELATED TO OUR TOPIC FOR TODAY.

THANK YOU